

YILDIZ TEKNİK ÜNİVERSİTESİ
BİLGİSAYAR MÜHENDİSLİĞİ BÖLÜMÜ



Ders: BLM2022 Bilgisayar Donanımı
ÖDEV-1 Raporu

Öğrenci Adı Soyadı: Sait Yalçın

Öğrenci No: 20011024

Öğrenci e-posta: sait.yalcin@std.yildiz.edu.tr

Ders Yürütücüsü: Dr.Öğr.Üyesi Ali Can KARACA

Nisan 2022

İÇİNDEKİLER

Soru – 1	3
Soru – 2	5

Soru 1

m_2 m_1 J k → Tam 3k lar için
 D_1 $\overline{D_1}$

D_1	q	q_1	J	k
0	0	0	0	X
0	1	0	1	X
1	0	1	1	X
1	1	1	0	0

D_1	q_0	q_1	q_2	q_3
0	X	X	X	X
1	X	X	0	X

D_1	q_0	q_1	q_2	q_3
0	0	0	X	X
1	X	1	X	X

q	q_1	J	k
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

m_2 m_1 q_3 q_2 q_1 q_0 J k J_0 k_0
 1 1 q_3 q_2 q_1 q_0 J_0 k_0 1 1

q_3	q_2	q_1	q_0	q_3	q_2	q_1	q_0	q_3	q_2	q_1	q_0	J	k	J_0	k_0
0	0	0	0	1	1	1	1	1	X	1	X	1	X	1	X
0	0	0	1	0	0	0	0	0	X	0	X	0	X	0	1
0	0	1	0	0	0	0	1	0	X	0	X	X	1	1	X
0	0	1	1	0	0	1	0	0	X	0	X	X	0	X	1
0	1	0	0	0	0	1	1	1	X	X	1	1	X	1	X
0	1	0	1	0	1	0	0	0	X	X	0	0	X	X	1
0	1	1	0	0	1	1	0	1	X	X	0	X	1	1	X
0	1	1	1	0	1	1	1	0	X	X	0	X	0	X	1
1	0	0	0	0	1	1	1	1	X	1	1	X	1	1	X
1	0	0	1	1	0	0	0	1	X	0	0	X	0	X	1
1	0	1	0	1	1	0	1	0	X	0	0	X	0	X	1
1	0	1	1	1	1	0	1	1	X	0	X	1	1	X	1
1	1	0	0	1	1	1	0	0	X	0	X	0	0	X	1
1	1	0	1	1	1	1	0	1	X	0	X	0	X	1	X
1	1	1	0	1	1	1	0	0	X	0	X	0	X	1	X
1	1	1	1	1	1	1	0	1	X	0	X	0	X	1	X
1	1	1	1	1	1	1	1	0	X	0	X	0	X	1	1

q_3	q_2	q_1	q_0
00	1	0	0
01	0	0	0
11	X	X	X
10	X	X	X

q_3	q_2	q_1	q_0
00	X	X	X
01	X	X	X
11	0	0	0
10	1	0	0

$$J_3 = \overline{q_2} \cdot \overline{q_1} \cdot \overline{q_0}$$

$$k_3 = \overline{q_2} \cdot \overline{q_1} \cdot \overline{q_0}$$

②

j_2, q_2, q_0

	00	01	11	10
00	1	0	0	0
01	x	x	x	x
11	x	x	x	x
10	1	0	0	0

k_2

	00	01	11	10
00	x	x	x	x
01	1	0	0	0
11	1	0	0	0
10	x	x	x	x

$$\Rightarrow \frac{j_2, k_2, 10}{\overline{q_1} \cdot \overline{q_0}}$$

j_1, q_1, q_0

	00	01	11	10
00	1	0	x	x
01	1	0	x	x
11	1	0	x	x
10	1	0	x	x

k_1, q_1, q_0

	00	01	11	10
00	x	x	0	1
01	x	x	0	1
11	x	x	0	1
10	x	x	0	1

$$\Rightarrow \frac{j_1, k_1}{\overline{q_0}}$$

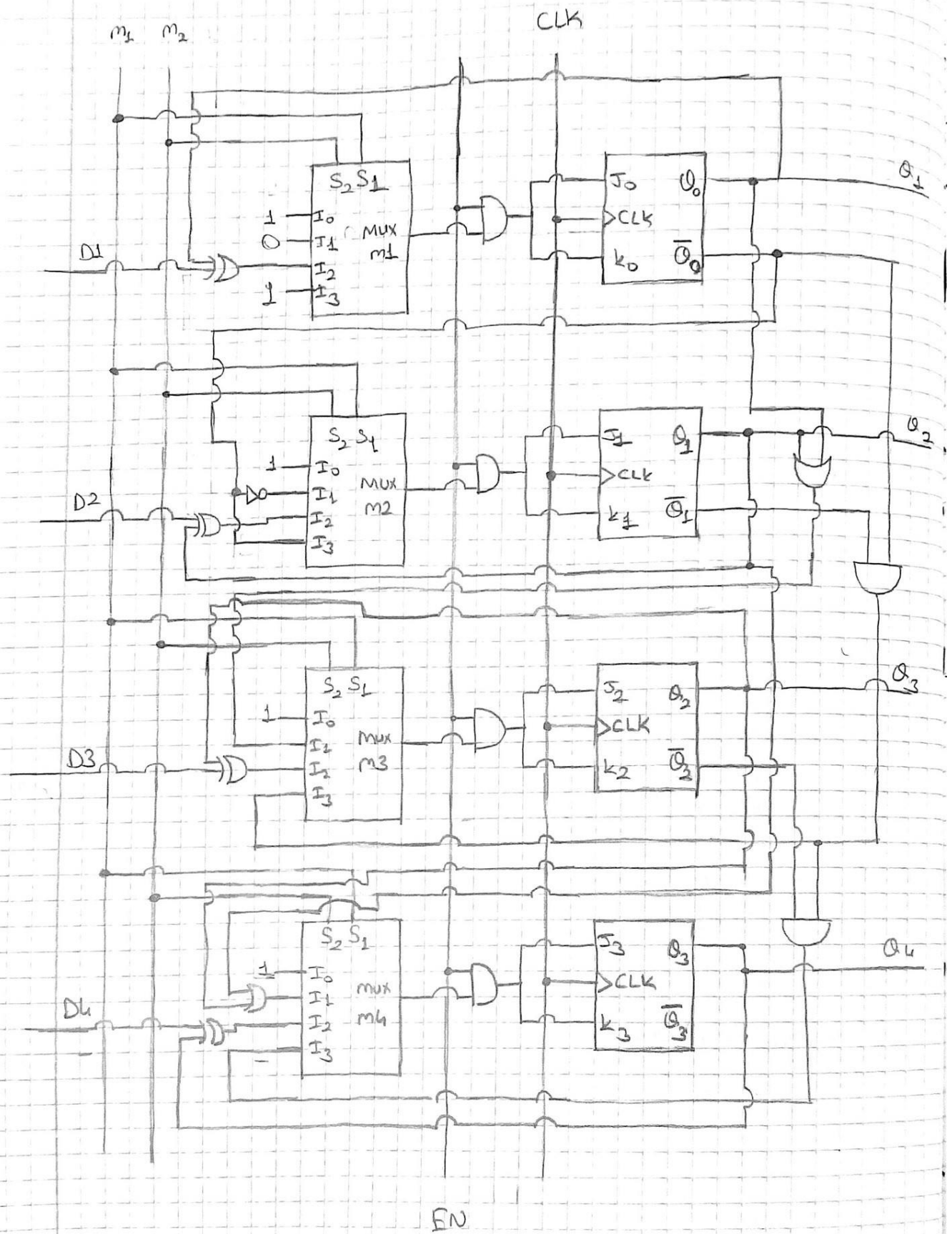
j_0

1	x	x	1
1	x	x	1
1	x	x	1
1	x	x	1

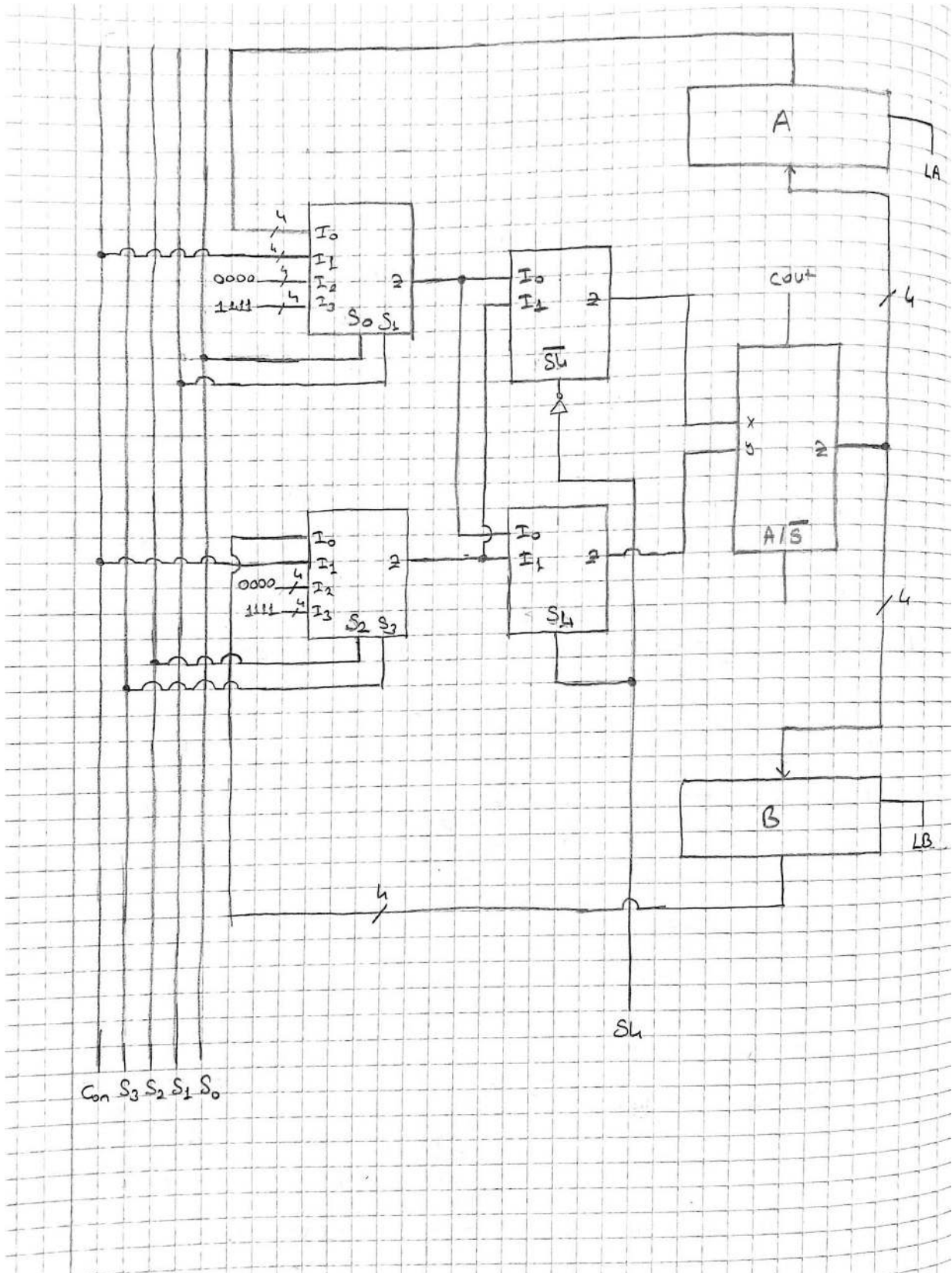
k_0

x	1	1	x
x	1	1	x
x	1	1	x
x	1	1	x

$$\Rightarrow \frac{j_0, k_0}{01}$$



Soru 2



Kontrol İşareti:

Not: Adder/Subtractor'ın (x-y) yaptığı varsayılıyor.

- $tar \leftarrow con$

$A \leftarrow con: S1=0, S0=1; S3=1, S2=0; S4=0; A/\bar{S}=1; LA=1, LB=0$

$B \leftarrow con: S1=0, S0=1; S3=1, S2=0; S4=0; A/\bar{S}=1; LA=0, LB=1$

- $tar \leftarrow Src$

$B \leftarrow A: S1=0, S0=0; S3=1, S2=0; S4=0; A/\bar{S}=1; LA=0, LB=1$

$A \leftarrow B: S1=1, S0=0; S3=0, S2=0; S4=0; A/\bar{S}=1; LA=1, LB=0$

$B \leftarrow B: S1=1, S0=0; S3=0, S2=0; S4=0; A/\bar{S}=1; LA=0, LB=1$

$A \leftarrow A: S1=0, S0=0; S3=1, S2=0; S4=0; A/\bar{S}=1; LA=1, LB=0$

- $tar \leftarrow A+B$

$A \leftarrow A+B: S1=0, S0=0; S3=0, S2=0; S4=0; A/\bar{S}=1; LA=1, LB=0$

$B \leftarrow A+B: S1=0, S0=0; S3=0, S2=0; S4=0; A/\bar{S}=1; LA=0, LB=1$

- $tar \leftarrow A-B$

$A \leftarrow A-B: S1=0, S0=0; S3=0, S2=0; S4=1; A/\bar{S}=0; LA=1, LB=0$

$B \leftarrow A-B: S1=0, S0=0; S3=0, S2=0; S4=0; A/\bar{S}=0; LA=0, LB=1$

- $tar \leftarrow Src+1 \quad (Con=1)$

$A \leftarrow A+1: S1=0, S0=0; S3=0, S2=1; S4=0; A/\bar{S}=1; LA=1, LB=0$

$A \leftarrow B+1: S1=0, S0=1; S3=0, S2=0; S4=0; A/\bar{S}=1; LA=1, LB=0$

$B \leftarrow B+1: S1=0, S0=1; S3=0, S2=0; S4=0; A/\bar{S}=1; LA=0, LB=1$

$B \leftarrow A+1: S1=0, S0=0; S3=0, S2=1; S4=0; A/\bar{S}=1; LA=0, LB=1$

- $tar \leftarrow Src-1 \quad (Con=1)$

$A \leftarrow A-1: S1=0, S0=0; S3=0, S2=1; S4=0; A/\bar{S}=0; LA=1, LB=0$

$A \leftarrow B-1: S1=0, S0=1; S3=0, S2=0; S4=1; A/\bar{S}=0; LA=1, LB=0$

$B \leftarrow B-1: S1=0, S0=1; S3=0, S2=0; S4=1; A/\bar{S}=0; LA=0, LB=1$

$B \leftarrow A-1: S1=0, S0=0; S3=0, S2=1; S4=0; A/\bar{S}=0; LA=0, LB=1$

$\text{for } \leftarrow \overline{\text{src}}$

$A \leftarrow \overline{A} : s1=0, s0=0; s3=1, s2=1; s4=0; A/\overline{s}=0; LA=1, LB=0$

$A \leftarrow \overline{B} : s1=1, s0=1; s3=0, s2=0; s4=0; A/\overline{s}=0; LA=1, LB=0$

$B \leftarrow \overline{B} : s1=1, s0=1; s3=0, s2=0; s4=0; A/\overline{s}=0; LA=0, LB=1$

$B \leftarrow \overline{A} : s1=0, s0=0; s3=1, s2=1; s4=0; A/\overline{s}=0; LA=0, LB=1$

$\text{for } \leftarrow \overline{\text{src}} + 1$

$A \leftarrow \overline{A} + 1 : s1=0, s0=0; s3=1, s2=0; s4=0; A/\overline{s}=0; LA=1, LB=0$

$A \leftarrow \overline{B} + 1 : s1=1, s0=0; s3=0, s2=0; s4=0; A/\overline{s}=0; LA=1, LB=0$

$B \leftarrow \overline{B} + 1 : s1=1, s0=0; s3=0, s2=0; s4=0; A/\overline{s}=0; LA=0, LB=1$

$B \leftarrow \overline{A} + 1 : s1=0, s0=0; s3=1, s2=0; s4=0; A/\overline{s}=0; LA=0, LB=1$